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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	KQ BuildLab
Selected Domain	Domain 1
Selected SDG Track (1–17)	SDG 8
Project / Solution Name	FutureNow
Name (Leader)	Lai Kai Qi
Email & Phone (Leader)	kaiqi1004@gmail.com & 01135118611
University Affiliation(s)	UNIMAS
Number of Team Members	1

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Artificial intelligence is changing the skills required across industries, creating challenges for graduates and working adults who need to remain employable in a changing labour market. Job seekers may struggle to identify which of their existing skills are transferable, which competencies employers currently require, and what training would help them pursue suitable career opportunities.

In Malaysia, this challenge is relevant to SDG 8, which promotes sustained, inclusive and sustainable economic growth, productive employment and decent work. Although labour-market statistics and online job listings provide useful information, users may find it difficult to translate these sources into practical, personalised career decisions. A job title alone does not explain whether a person is prepared for a role or which skills they should develop next.

FutureNow addresses this gap by connecting labour-market information with individual skill profiles. Its intended users are university graduates and working adults seeking entry-level opportunities, career progression or career transitions. The project will use official Malaysian labour statistics and a curated sample of job listings to explore skill demand and identify gaps between users' current capabilities and selected job requirements.

By making career guidance more evidence-informed and actionable, FutureNow aims to support better training decisions, improve career readiness and contribute to more inclusive access to employment opportunities. The prototype will present its data coverage and limitations transparently rather than claiming to predict the entire Malaysian labour market.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

FutureNow is a planned AI-assisted career guidance application designed to help Malaysian graduates and working adults understand their employability and prepare for changing job requirements.

Users will be able to create a skills profile through a questionnaire or structured input for skills extraction. They can explore predefined career roles or paste a job description to identify required competencies. The system will compare the user's skills with the selected role's requirements, present an explainable matching result, highlight skill gaps and recommend a personalized learning roadmap.

The application will also present a Malaysian labour-market intelligence section using official statistics from the Department of Statistics Malaysia (DOSM) and a curated collection of job listings from MYFutureJobs. These sources serve different purposes: DOSM provides broader labour-market indicators, while job listings offer a limited snapshot of employer-stated requirements. A scenario-planning feature is intended to illustrate moderate and faster AI-disruption scenarios as exploratory estimates, not certain predictions.

AI will be used where appropriate for interpreting job descriptions, extracting and organising skills, explaining career matches and generating contextual learning recommendations. Structured data and transparent matching rules will support consistency and traceability. The team plans to apply skills developed through the SDC workshops in generative AI integration, database design and cloud deployment to build a functional, accessible web prototype.

FutureNow aims to make career planning more understandable and actionable while clearly communicating data limitations. Its recommendations will support, rather than replace, users' own judgement and professional career advice.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

FutureNow is developed as a web application using the following technology stack:

- **Frontend:** TypeScript-based web application for the user interface and career-planning workflow.
- **Backend:** Node.js 24 and Express 5 for server-side API functionality.
- **Data validation:** Zod, where used by the application.
- **Database:** PostgreSQL with Drizzle ORM, subject to confirmation of the active database connection in the submitted deployment.
- **Development and deployment:** Replit, with the source code maintained in a public GitHub repository.
- **Labour-market data:** Selected DOSM statistics and a curated sample of MYFutureJobs job-posting records, subject to confirmation of which datasets are integrated into the running prototype.
- **Career matching:** Structured skill data and explainable matching logic. Any generative-AI model or external API will be declared only if it is actually integrated into the application.

System block diagram

User → FutureNow web interface → Skills profile and job-role selection → Career-matching logic → Match results, skill gaps and learning recommendations.

Labour-market data → Data processing and presentation → Workforce insights.

Replit → Development, testing and prototype deployment.

Official SDC sponsor-provided resources

Replit Core access was provided through the SDC hackathon arrangement, according to the team's information. It is used to support development, testing and prototype deployment. The exact sponsor attribution and any additional sponsor-provided API or cloud credits will be confirmed against the organiser's instructions.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

FutureNow follows a user-centred design approach focused on the career-planning challenges faced by graduates and working adults.

Empathise: Identify difficulties in understanding employer skill requirements, evaluating transferable skills and choosing suitable training.

Define: Frame the problem as a gap between an individual's current capabilities and the requirements of changing career opportunities.

Ideate: Design an integrated workflow combining skills profiling, job-description analysis, explainable career matching, skills-gap identification and learning recommendations.

Prototype: Build a functional web application with a focused end-to-end workflow before expanding secondary features. Malaysian labour-market indicators and collected job listings will be presented separately to avoid confusing national statistics with a limited sample of vacancies.

Test and iterate: Conduct usability tests using representative tasks, such as creating a profile, comparing it with a job role and identifying the next learning steps. Evaluate task-completion rates, recommendation relevance, response time and data traceability.

The prototype will be deployed through Replit for demonstration. Testing will inform interface improvements and help identify technical limitations before the final presentation. Scenario-planning outputs will be labelled as estimates rather than established forecasts.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	**UN SDG Target 8.5** Achieve full and productive employment and decent work for all, including young people. **UN SDG Target 8.6** Reduce the proportion of young people not in employment, education or training. FutureNow supports these targets indirectly by helping users identify relevant career opportunities, understand skill gaps and plan further learning. The prototype will measure career-planning usability and recommendation quality; it will not claim to directly reduce national unemployment or the NEET rate.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	At the start of the prototype evaluation, FutureNow has no established usability-test baseline. The initial task-completion rate is therefore recorded as not yet measured (baseline pending initial user testing) rather than assuming a zero-percent success rate.

Metric Category	Your Response
	The team will establish the baseline by asking test participants to complete the predefined career-planning tasks without assistance. The results will be documented before further usability improvements and compared with the Phase 2 target of at least 80% unassisted task completion. The broader problem is the difficulty job seekers may face in translating labour-market information and employer-stated skill requirements into practical career and training decisions. National employment indicators will be cited separately as context, not as evidence that FutureNow itself has reduced unemployment.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Percentage of usability-test participants who successfully complete the core career-planning workflow. **Measurement:** (Number of participants completing all predefined core tasks without assistance ÷ total participants tested) × 100. Core tasks include creating a skills profile, obtaining a career-match result and identifying at least one relevant skill gap or learning recommendation. Additional indicators will include recommendation relevance, response time and the proportion of displayed job records with traceable source information.

Metric Category	Your Response
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Achieve a minimum 80% unassisted task-completion rate among usability-test participants by the final prototype presentation. The target will be evaluated through predefined tasks and documented test results. The number of participants, task definitions and completion criteria will be reported alongside the percentage. This is a prototype usability target, not a claim of an 80% improvement in employment outcomes or a reduction in national unemployment.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- [ ] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- [/] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- [ ] The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- [/] All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes / No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

/ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	**Type (code/dataset/model)**	**License**
TypeScript	Code / language tooling	Apache-2.0
Node.js	Runtime	MIT
Express 5	Open-source framework	MIT
Drizzle ORM	Database library, if included in the submitted application	Verify against the installed package licence
Zod	Validation library, if included in the submitted application	MIT
DOSM labour-market datasets, where used	Public dataset	Applicable DOSM data-use terms; verify the specific source
MYFutureJobs job-posting	Collected source records	MYFutureJobs terms; reuse

records, where used		and redistribution permissions must be checked

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes / No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



[Lai Kai Qi]

25/10/2026